

Chapter 3:

3.0 System Analysis

3.1 Introduction

Analysis is an important intermediate stage between investigation and design. It involves examining all the facts collected or gathered in order to make a proper assessment of both the proposed and existing system. The aim of this stage is to ensure actual user requirements are obtained. These requirements are most often gathered from the users of the system.

For analysis to be achieved different methods may be used to collect facts or data from the users. These methods are **interviews, observation and questionnaires**. During this analysis interviews and observation were used to gather data from users of the existing system and those having systems with similar specifications as the proposed system. I also collected some secondary data; the major source was literature from the books of the organization. Here I got to understand the operations of the existing system.

For the interview the major source was the users who have directly been involved with the website. The responses I got included:

- Website could not provide a single function via a page.
- The website was unfriendly and not interactive as users expected.
- It is difficult for users to customize their own pages.

The findings for the investigation were:

- Information could not be accessed in a diverse way
- The website does not provide a single function via page as well as point of access to information on World Wide Web.

a) Possible solutions to the organization problem

This section shows the comparison of the possible options to the business problems. Possible solutions are compared to each other with respect to cost, level of application risk and efforts required.

Table 1.0

Options	Cost of purchase in Kshs.	Level of risk	Effort Required
Purchase off the shelf software(Web portal)	Approx. 160,000	Low- Well –designed solution and industry	Medium - Simple buy and install

		approved. But can still fail if not implemented properly.	
Build Own	Approx. 30000	Medium- well designed solution, developer has no much experience	High

3.2 Feasibility Study

The purpose of this section is to define the problem and decide whether or not a new system to replace an existing one is viable or feasible. This section will cover the following activities:

- Assessing of the various areas of feasibility study. Which include;
 - technical feasibility,
 - social feasibility,
 - economical feasibility,

3.2.1 Technical Feasibility

This is a type of study that deals with equipment and software. The specifications for the hardware and software are as follows:

- ✓ A total of 10 computers are required. They should have a capacity of 40 GB and above, processing speed of 0.8GHZ and memory of not less than 256 MB.
- ✓ A LAN network that is to support a client/server architecture.
- ✓ The operating system preferably windows XP.
- ✓ Anti virus software to protect against harm to data in the computer.
- ✓ MySQL for the DBMS.
- ✓ Apache for server side scripting
- ✓ Perl, HTML,perl modules for the application
- ✓ Server with at least 2GB RAM,duo core processor

Training of portal administrators is required since they are the core people of the web portal. They can be taught through use of a tutor or through reading tutorials that can assist them to know more about operation of the web portal.

A network administrator is needed to install and maintain the Local Area Network that will allow sharing of data among different computers.

3.2.2 Social Feasibility

It deals with the effects of the system on the current society within the system. It is also known as operational feasibility. The users of the system appreciate the proposed system. For direct users it will enable them to have such as e-mail, news, stock prices, infotainment, and other features at their click of mouse. Students do not need to pick forms at the university; they get any kind of information whenever they are as long as they are connected

3.2.3 Economical Feasibility

This is aimed at determination of whether or not to continue with the project depending on whether the project is economically viable. It is determined by use of a cost benefit analysis. The proposed system will lead to reduced expenses.

a) Costs

The costs of the proposed system include:

- Equipment costs – Costs incurred when buying computers, data cables and networking devices.
- Software cost- Costs incurred when buying software e.g. MySQL 5 or above and Windows XP.
- Development costs – Costs incurred during the development of the system.
- Operational costs – costs incurred during the installation of the network devices and computer software.

b) Benefits

- Reduces operational expenses e.g printing costs, papers since most information that was once being printed can now be posted on the web portal
- Students save time visiting the administration or the university just to check notices e.g opening/closing dates, results and other important information the students need to know.
- Users are saved from fatigue of filling and following through large chunks of paper as they look for details.

- No time wastage in exchange of information as it is always available hence saves on cost and energy e.g students can consult about their assignments in through forums and discussions.

3.2.4 Feasibility Study Report

3.2.4.1 Introduction

The alternative proposed system(web portal) caters for the overhead costs incurred through; allowing sharing of data among various departments hence no need to follow up on the information through making calls, Ensuring accuracy and integrity of data hence not susceptible to human errors, does not require much human effort hence saves students and staff from fatigue

3.2.4.2 Recommendation

Even though there are initial costs or capital costs required for the success of the proposed system, the benefits in terms of availability, accuracy, convenience and currency out do costs. The proposed web portal will operations, students satisfaction on how information is accessed compared to the normal website.

3.3 Software Requirement Specification

This part will cover areas that will help in the complete understanding of the system and the environment that it is going to be subjected to.

3.3.1 Introduction

This section introduces the SRS describing its purpose and scope and defining all terms, acronyms and abbreviations required to properly interpret the system.

3.3.1.1 Purpose

The purpose of this SRS is to help; users to adequately and accurately describe what they wish to obtain and programmers to understand exactly what the customers want.

This SRS is intended not only to be used by direct users and programmers but also the users who are indirectly impacted by the proposed system like the financiers who fund the proposed system.

3.3.1.2 Scope

The web portal is the software product that is being analyzed in this SRS. The software product will be used to achieve the following tasks:

- Search and navigation by users.
- Information integration.
- Personalization.
- Notification services.
- Collaboration and groupware.

3.3.2 Product functional and non-functional requirements

3.3.2.1 Functional Requirements

This section of the requirements states the function of the system i.e. what the system should do.

a)User Authentication

All users must identify themselves using an E-mail address and a password in order to gain access to more features. However the web portal allows users to access general information without necessarily having to have been registered.

b)Search and Navigation

The portal should be able to support search and navigational services to all users

c)Personalization

Here the users should get information specifically tailored for his needs. Personalization is done based on user roles.

d)Notification

Users should be able to get information automatically from the server e.g information concerning password expiry and other forms of notification.

e)Collaboration and groupware

Right person are brought together with right people

f)Information Intergration

A portal should warrant the integration of information from disparate sources.

3.3.2.2 Non-Functional Requirements

These are requirements that stipulate the performance levels required of the system for various kinds of activities to be performed.

a)Performance

The web portal should be efficient and fast in access of information requested by users.

b)Reliability

The web portal should have some tolerance to some of the errors that may occur. For example entry of invalid values by users should not make the portal to halt rather it should provide error messages trying to inform the user of the error made.

c)Availability

The user should be able to access the web portal at any point in time and also access and be able to manipulate the exact details he/she needed.

d)Usability

The web portal should have an easy to use and easy to understand user interface. Also the error messages should consist of simple language. This is so as to ease the operability of the system.

3.3.2.3 User Characteristics

The intended users (students, staff) should possess the following qualities.

- They should be computer literate.
- Have an understanding of navigations.

3.3.3 Specific Requirements

This section of the SRS contains all the requirements to a level of detail sufficient to enable designers to design a system to satisfy those requirements. Below is a brief description of the entities that participate in the web portal operation

- **Users** - They include students, JKUAT staff. They are identified by email address and name.
- **Portal Administrators**— They are like the technical staff. They define rights for users, upload portal with material and monitor events, add gizmo tags(classes of custom code whose sole purpose is to include dynamic content to your site).
- **Default users**—They are group of users who do not need to have registered, and they are allowed to access basic information on the web portal.

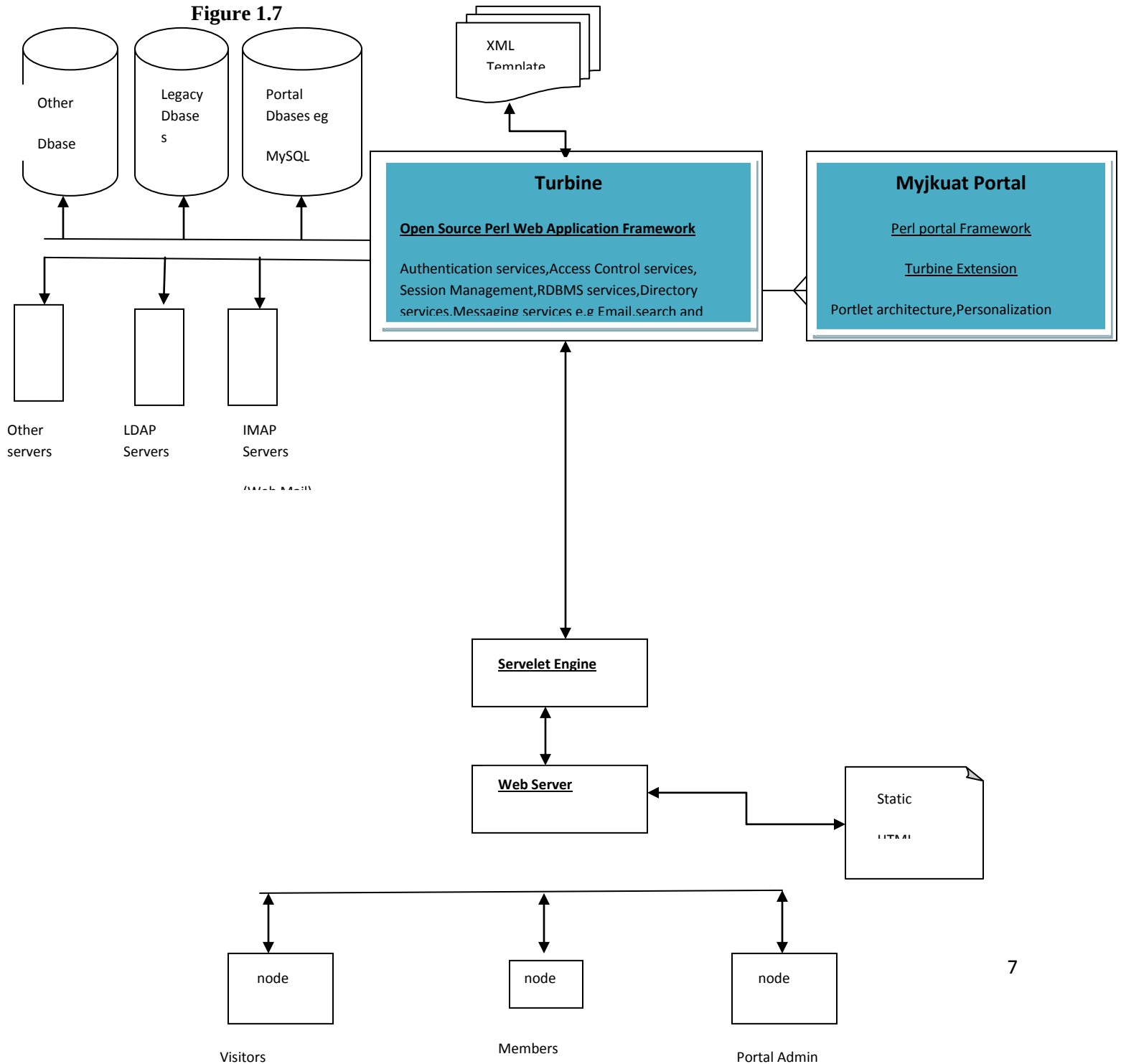
Chapter 4: System Design

This section explains design issues involved in the system architecture, interface designs of stand alone application and Database modeling.

3.4.1 Portal Architecture

System Architecture for the proposed web portal is based on the open source perl for web applications. The servlet engine include the Apache as well as web server.

Figure 1.7



3.4.2 Recommended Channels

It is a representation of the web pages in the web portal and their sections. The channels are a visual representation of the pages available on the web portal and what they do

Figure 1.8 General.

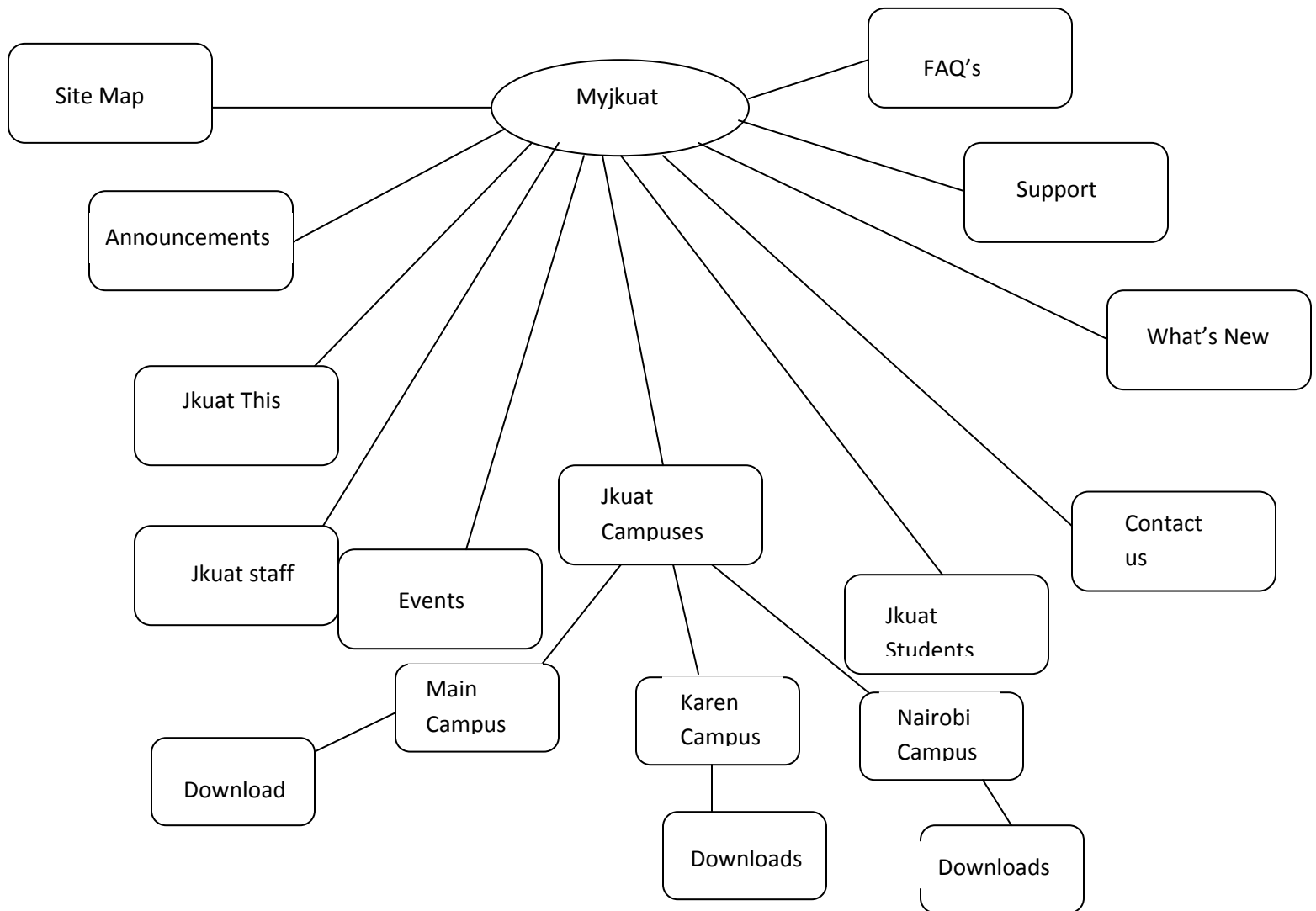


Figure 1.9 :Events/student/Announcements

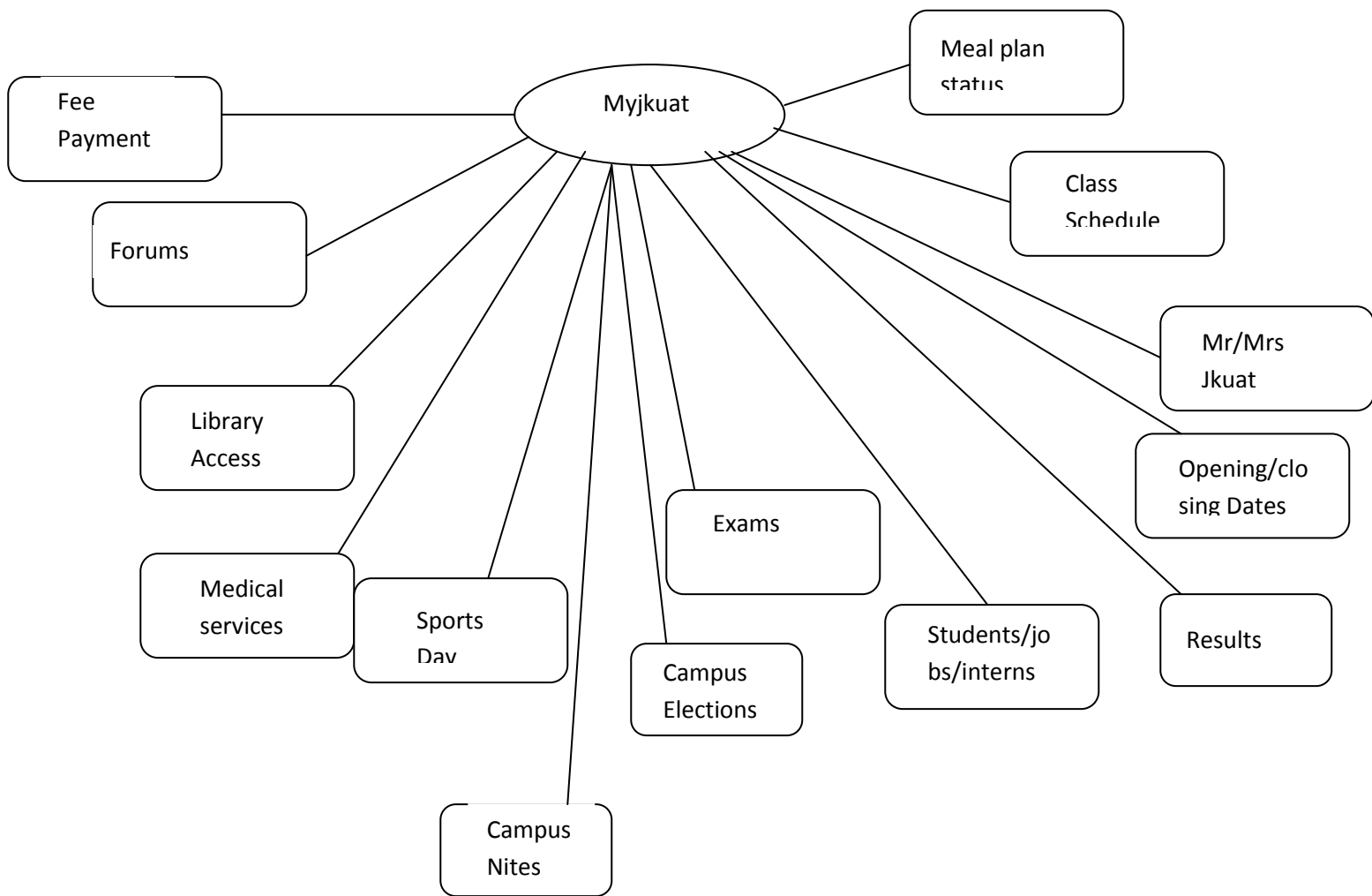


Figure 2.0 : staff/Jkuat This Week

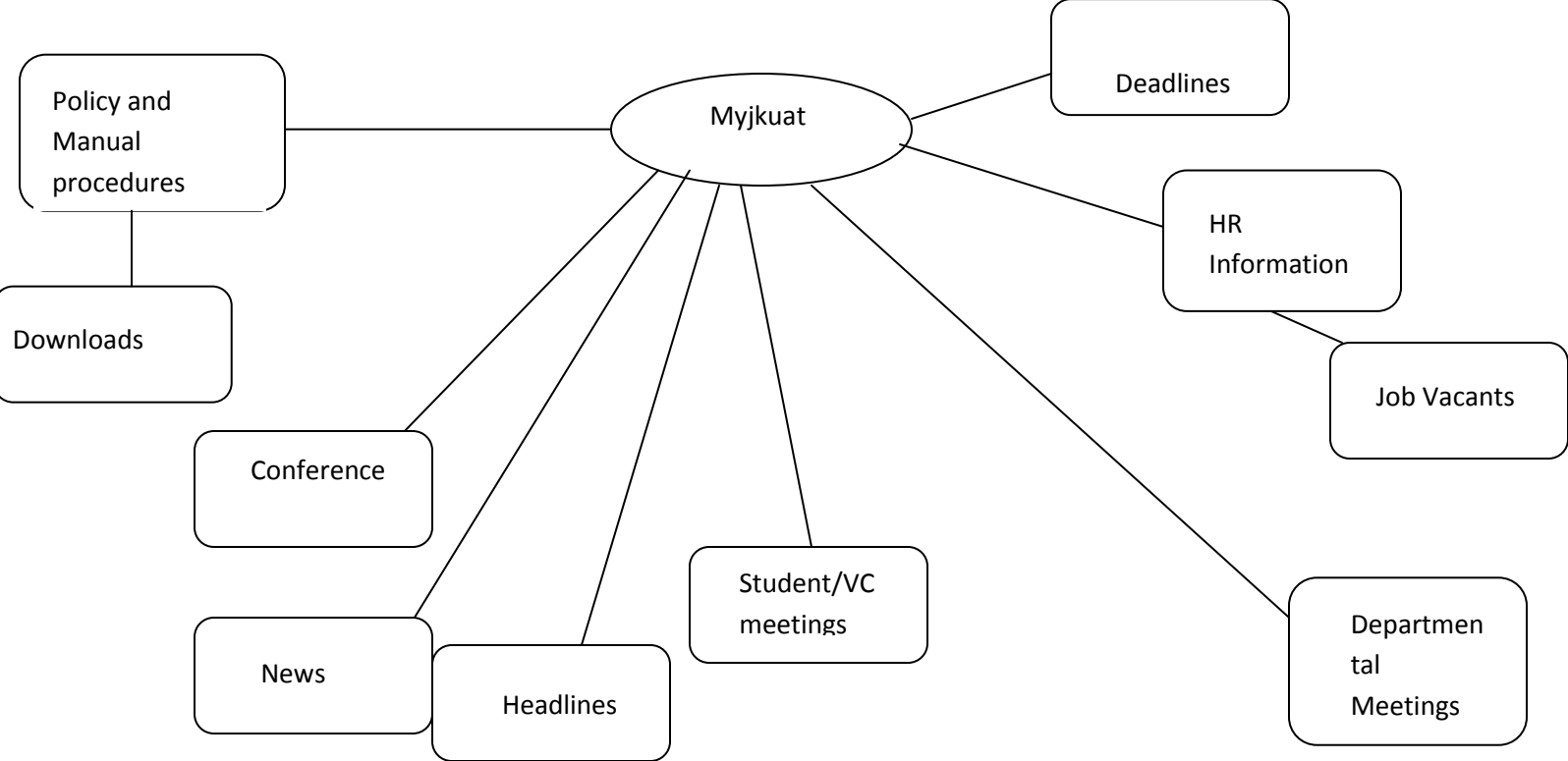
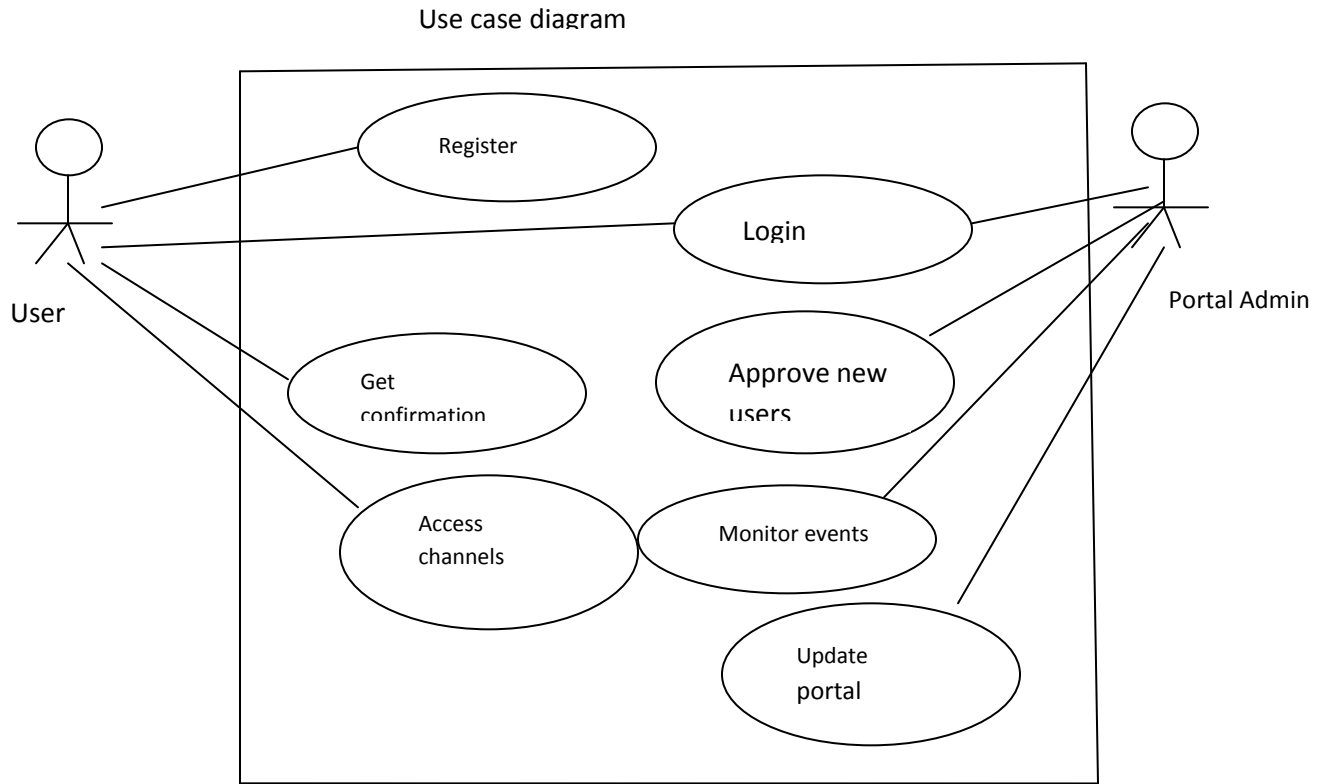


Figure 2.1: Use case Diagram



3.4.3 Data Flow Diagrams

It is a representation of the properties of the data within an existing or proposed system. It involves shifting fact collected during the investigation state process into a data model concept. The objects, relationships and characteristics from the investigation stage are represented in a graphical notation.

They provide an effective form in which to represent the movement of data through a system and the associated transformations of the data resulting from various processing actions taken upon it.

a)Context Diagram

Figure 2.2 :Context Diagram

Fig : Context Diagram

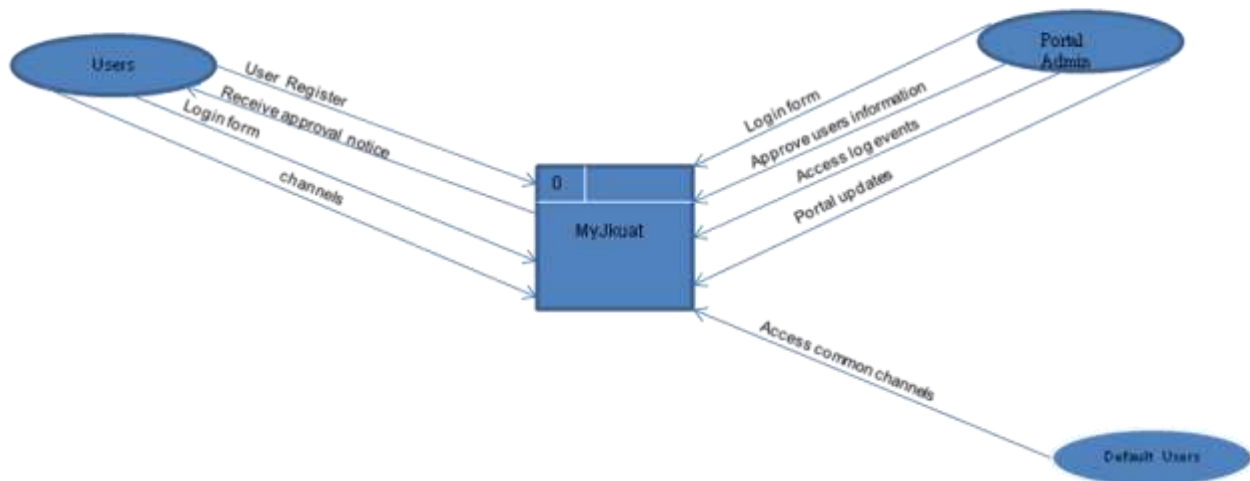


Fig 2.3 Diagram Zero

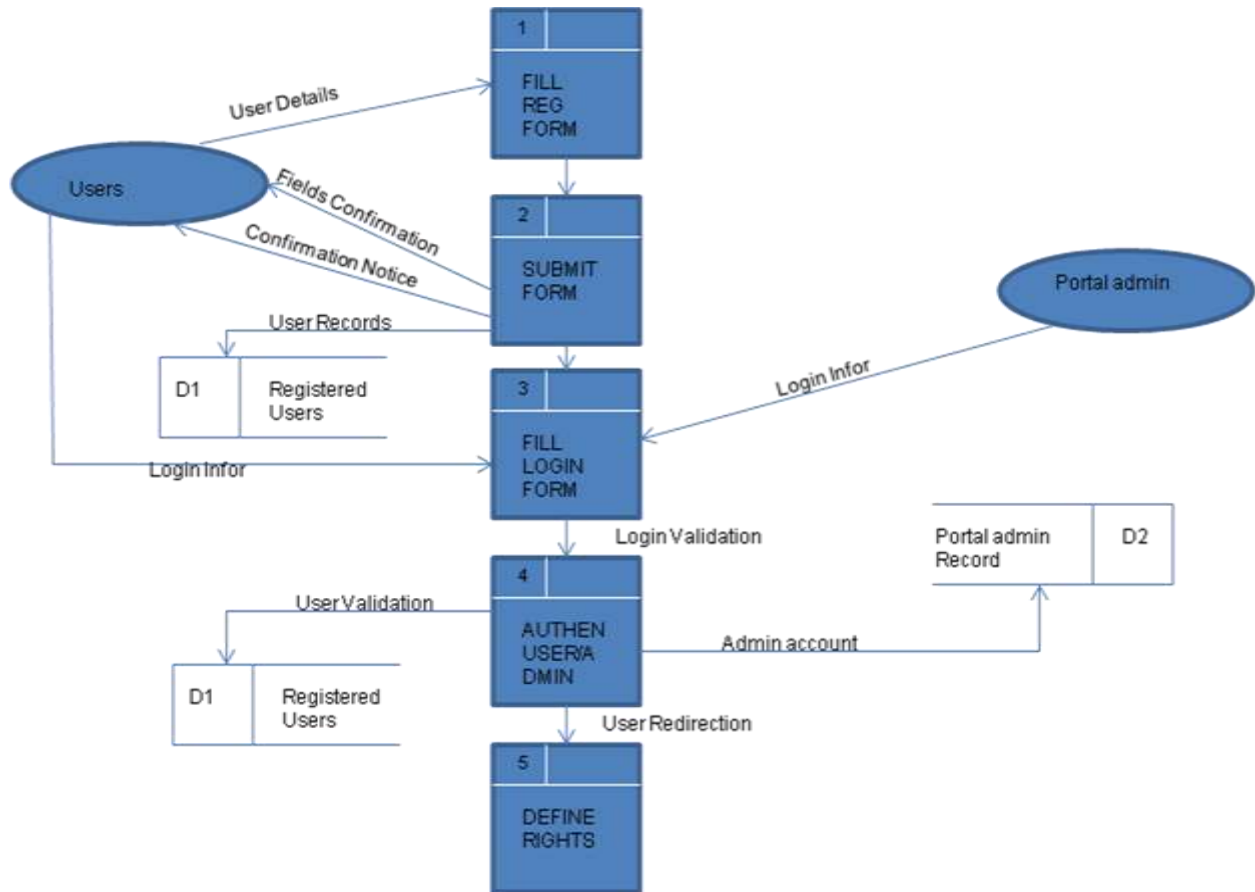


Fig: Diagram Zero(0)

Figure 2.4 Diagram 1 (a)Submit Form

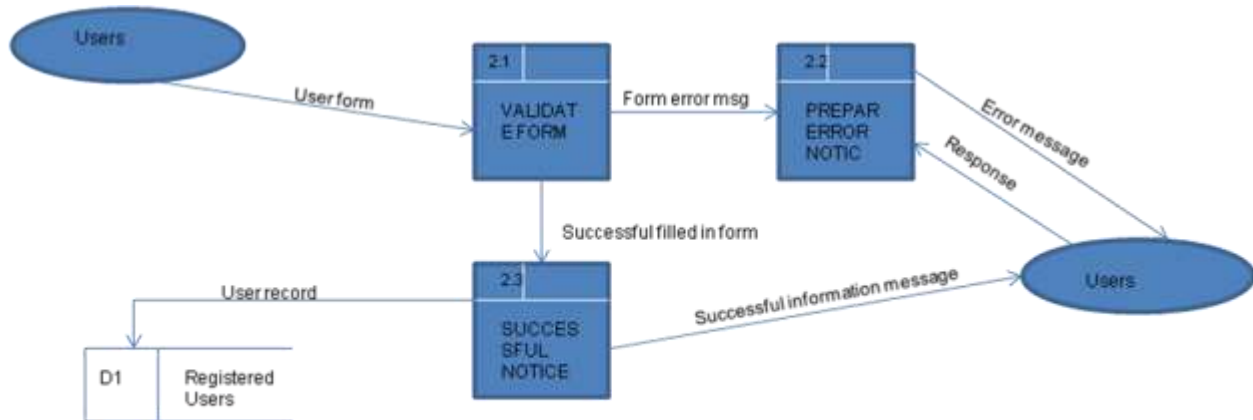


Fig: Diagram 1 (a)Submit Form

Fig 2.5 Diagram 1(b)Authenticate User

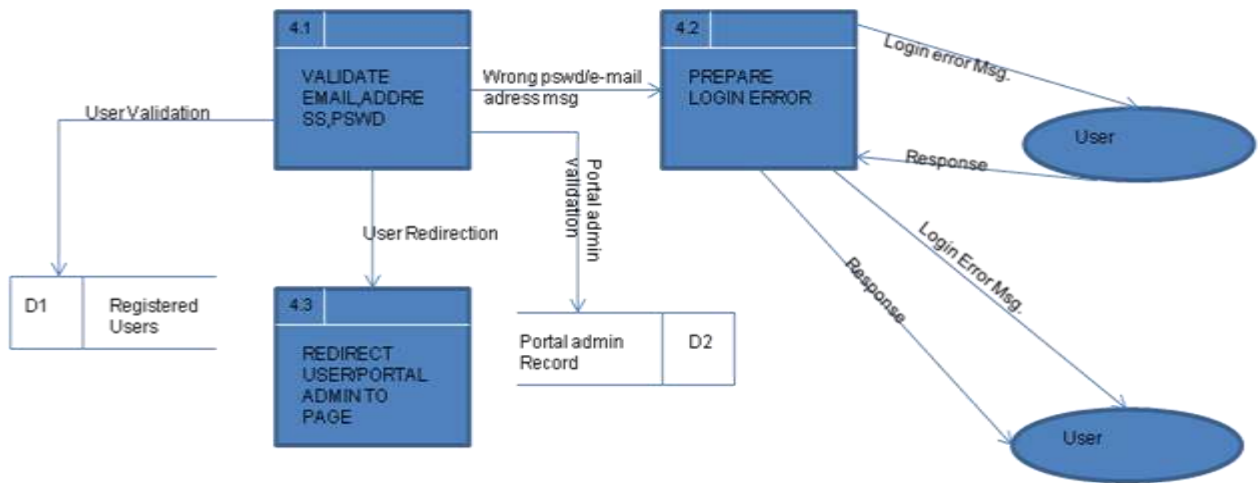


Fig:Diagram 1 (b)authenticate User

Fig 2.6: Diagram 1(c) Define access Rights

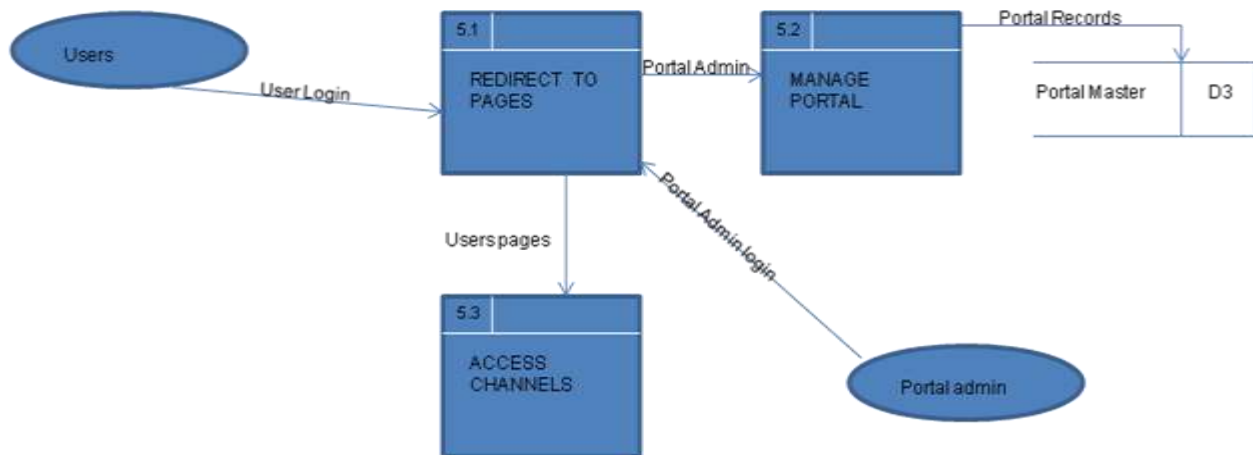


Fig :Diagram 1 (c) Define Access Rights

Fig 2.7: Diagram 2 (c) Manage Portal

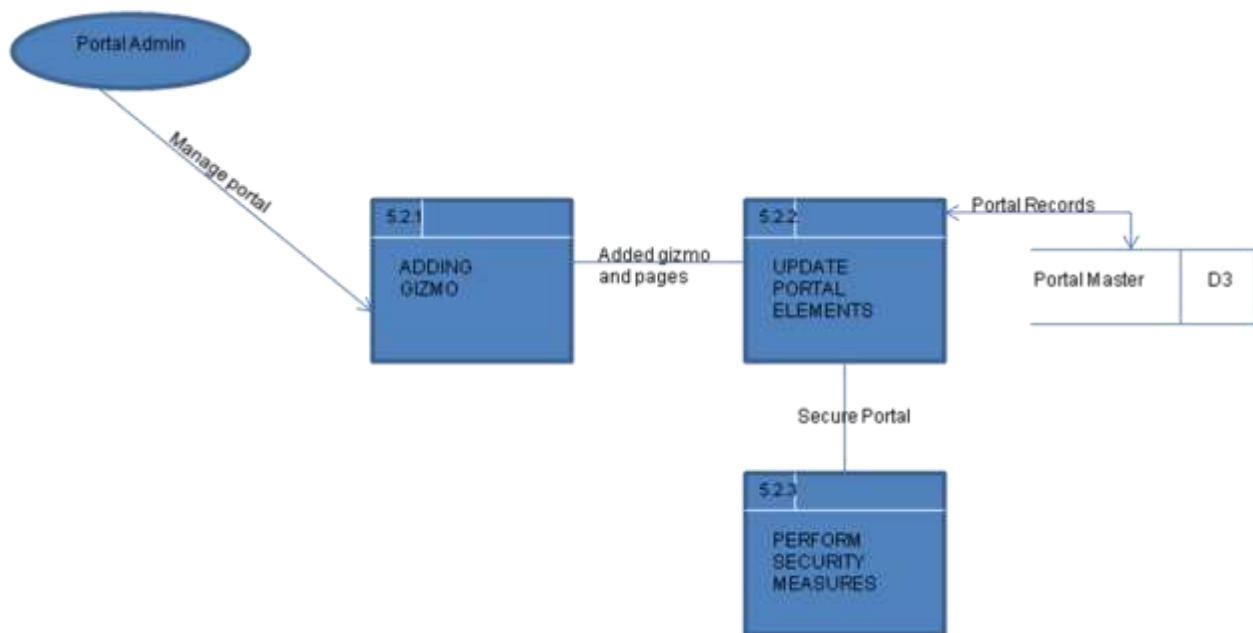


Fig :Diagram 2 (c) (l)Manage Portal

Fig 2.8: Diagram 2 (c) (ii) Access channels

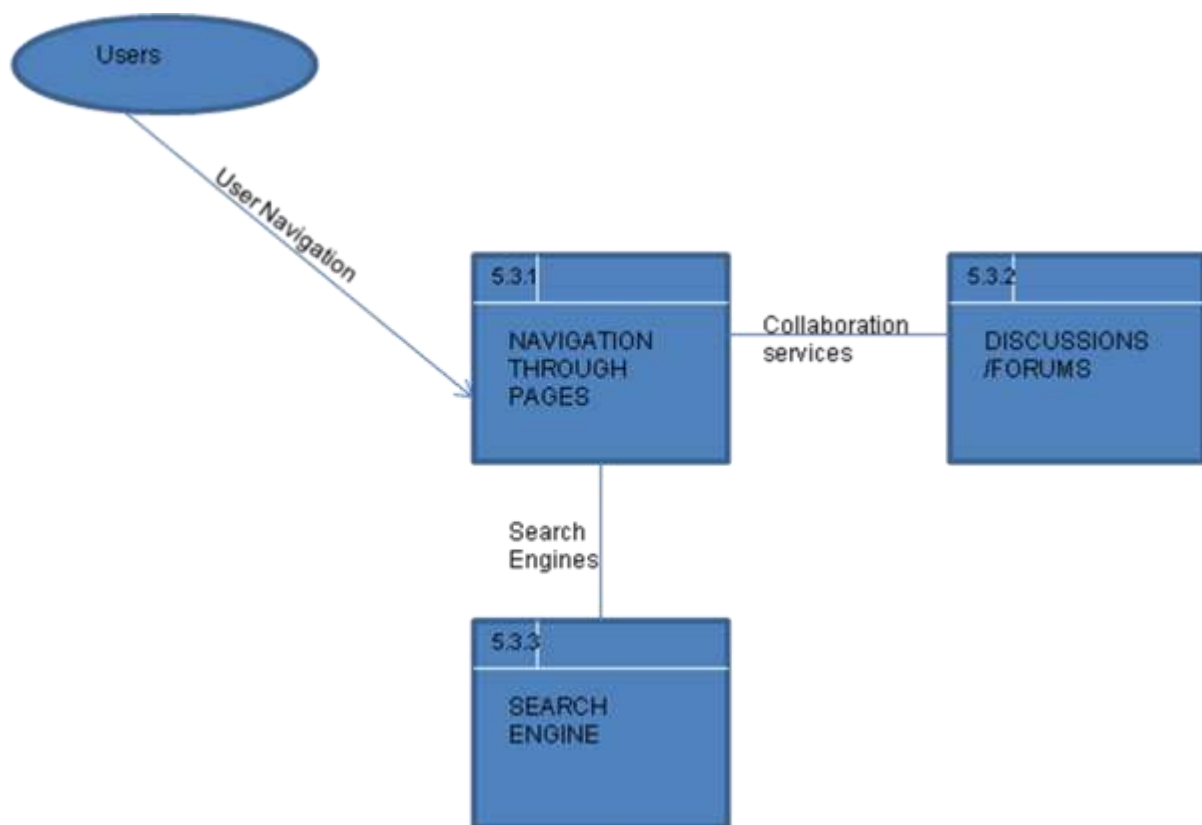


Fig :Diagram 2 (c) (ii) Access Channels

Fig 2.9: Diagram 3(a) Perform Security Measures

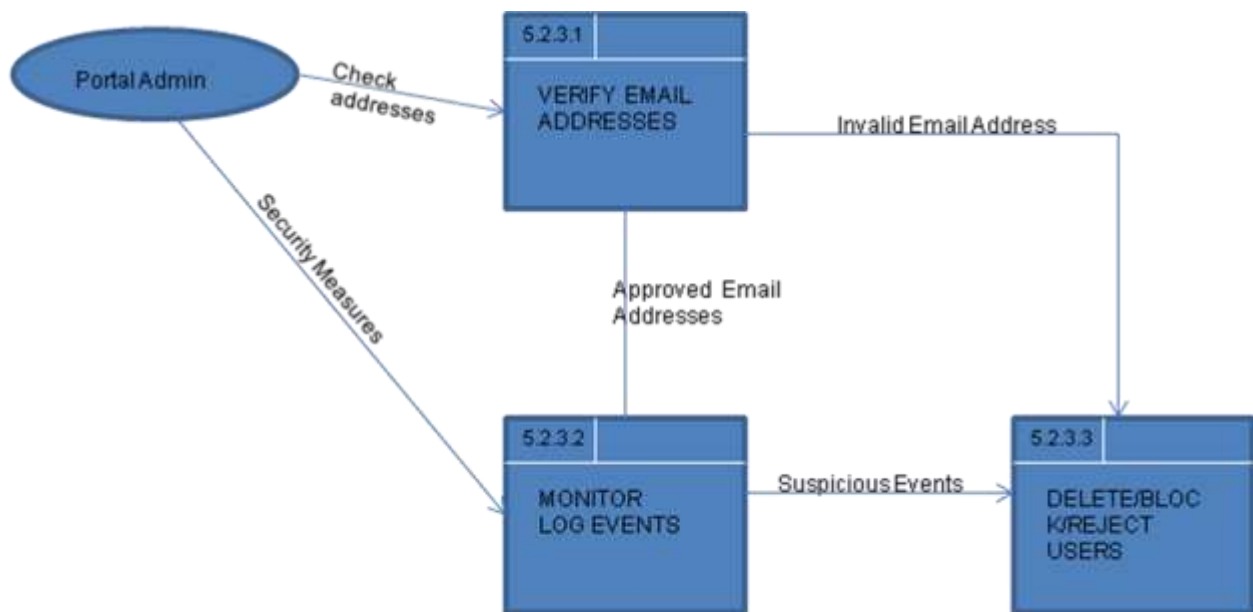


Fig: Diagram 3 (a)Perform Security Measures

3.4.4 Data Dictionaries

Diagram Zero.

User detail= user Email + Personal information + password

Field confirmation status= Dialog box + Error field location + Solution

Confirmation status= Dialog box + Confirmation message

Completed form= user email address + password + personal information

User record= user email address + password + other personal information

Submitted form= fully completed form

User login information= Email address + password

Portal Admin login information= Username + password

User redirection page= WebPages +Gizmo +events + Discussion

Diagram 1(2.submit form)

Form error notice= Dialog box +error message

Response= Refilled form + corrections

Successful form= dialog box + successful message.

Successful information notice= dialog box + awaiting approval message

Diagram 2(5.2 Manage portal)

Manage portal= access rights + editing mode

Added gizmo= gizmo name + description

Portal record= gizmos +users information + skins +Databases + image library

Secure portal= log events + user management + user parameters + system parameters.

Diagram 2(5.3 Access channels)

User navigation= site map + pages + sections + gizmos

Collaboration= discussion + forum name + members

Search activities= items to searched

Diagram 3(5.2.3 perform security measures)

Invalid email= wrong email address + wrong password

Suspicious events= irrelevant discussions + mimicking sites

Security measures= monitor portal events + granting rights and withdrawing.

Key

= Composed of

+ and

3.4.5 Data structure

User,s Name	Page	Gizmo
Firstname*	PageName*	GizmoName*
Middlename	Description	Description
Lastname*	KeywordSynonym	Attachment
Nickname		KeywordSynonym
EmailAddress*		
Password*		

Portal admin

Username*

Password*

Key

*-Mandatory Field

3.4.6 Data structure Description

User's Name	Page	Gizmo
=Firstname + (Middlename) + Lastname + (Nickname)+ EmailAddress+ Password + (HomePage)	=PageName + Description + (KeywordSynonym)	=GizmoName + (Description) + (Attachment) + (KeywordSynonym)

Portal admin

Username +
Password

Contacts

=(address line)
(city)
(state)
(zip code)
(country)
(telephone)
(mobile)
(fax)

Key

+ and

() Optional entry or field

= Composed of

3.4.7 Database Design

In this section database designing for the proposed system is explained. The methodology for designing database consists of three phases that is conceptual design, logical design, and physical design. These three phases are explained as follows:-

Conceptual Database Design – conceptual database design includes the identification of the entities, relationship between entities and defining the attributes.

Logical Database Design – this level helps in designing the relations and constraints.

Physical Database Design – this level includes the physical implementation of relations into database management system.

Entity-relationship Diagram helps in describing the entities and relationship. The database designing was done from beginning and during database designing it is ensured that Database model is in the normalized form. The database design for the organization is based on the following ER Diagram showing entities and their relationships.

3.4.7.1 Conceptual Database Design**4.1.1 Entity Relationship Modeling**

It is a detailed logical representation of data for an organization. It is expressed in terms of entities, relationship between or among the entities and attributes of both the entities and relationships.

The following is a composition of the entities and attributes used in the system:

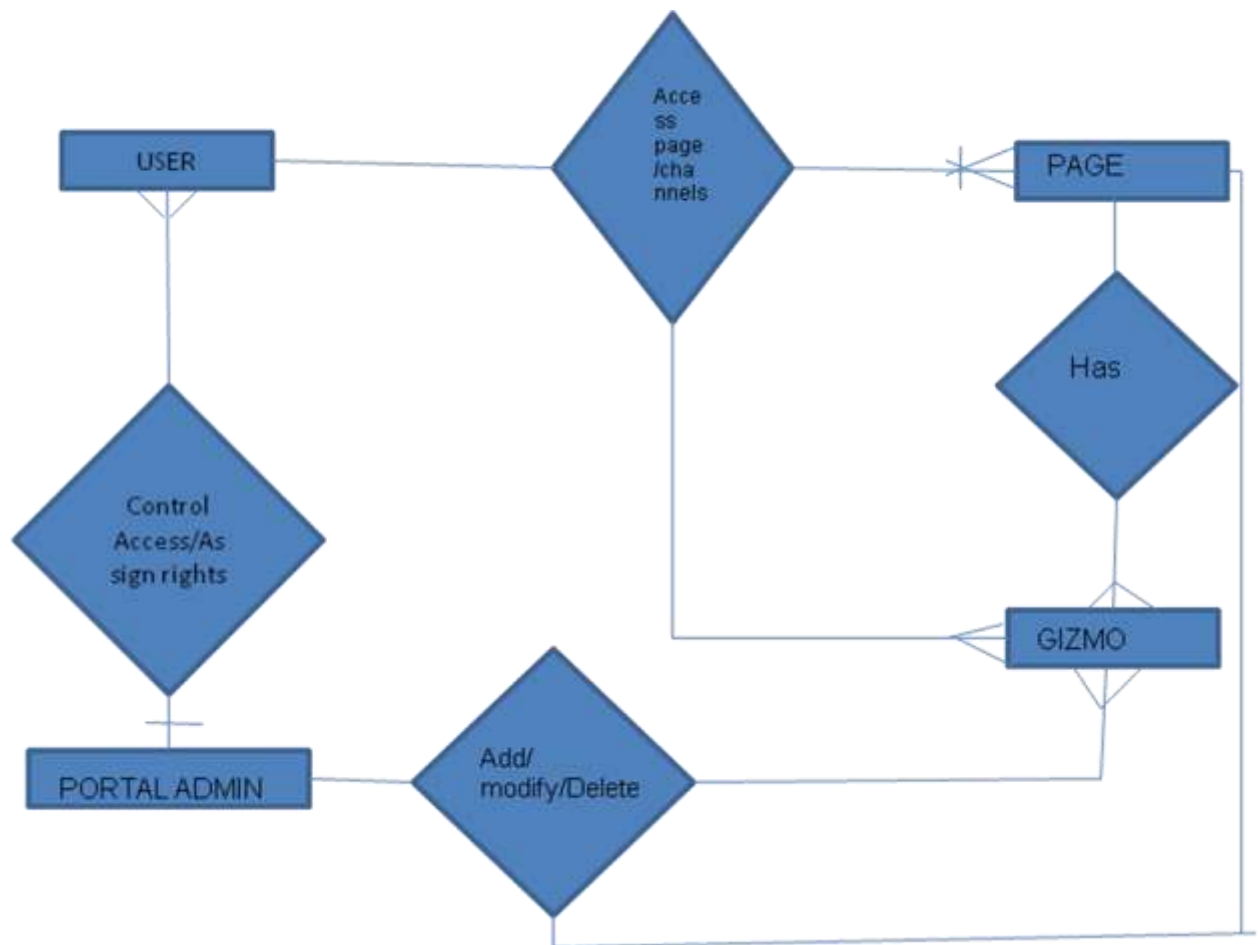
USER=[**Emailaddress**, FirstName, LastName, Password, Contacts]

PAGE=[**PageName**, Description, KeywordSynonym]

GIZMO=[**GizmoName**, Description, Attachment, KeywordSynonym]

PORTAL_ADMIN=[**Password**, UserName]

Figure 3.0 ER Diagram



3.4.7.2 Logical Database Design

User table

Table 1.1

Attribute name	Data type and length	Validation
<u>Email address</u>	char	Unique primary key
FirstName	varchar	
LastName	char	
Password	varchar	
Contacts	varchar	

Page Table

Table 1.2

Attribute name	Data type and length	Validation
<u>PageName</u>	char	Unique primary key
Description	char	
KeywordSynonym	char	

Gizmo Table

Table 1.3

Attribute name	Data type and length	Validation
<u>GizmoName</u>	char	Unique primary key
Description	char	
Attachment	char	
KeywordSynonym	char	

Portal Admin Table

Table 1.4

Attribute name	Data type and length	Validation
Username	char	Unique primary key
Password	varchar	

Chapter 5:

4.0 System Implementation

This chapter elaborates the Implementation issues of how the system design was implemented in terms of database implementation; application implementation for presentation or the user interface.

Implementation involves the co-ordination of the user department and the IT department in getting the new system in operation. Planning for it begins early in the design stage. It covers the following areas:

- a) Training of staff
- b) Programming
- c) System testing
- d) Changeover procedure
- e) Review and maintenance

4.1 Database Implementation

This section elaborates the issues regarding the designing of database for the proposed system. As stated earlier database was designed using MySQL 5.0. Designing of the database for the system involved identifying the entities, their relationship and constraint whereas implementation of database is completely different from database design; database designing is concerned with “*what*” whereas database implementation is more concerned with “*how*”.

4.2 Interface Implementation

This section elaborates structure of the interfaces, designed for the standalone Application. Actual interfaces were developed on personal computers. Standalone applications were designed using Perl open source framework. It provides functionalities to create interfaces using drag and drop facility that helps in easy development of powerful applications.

4.2.1 User Interface Design

In this section user interface designing issues will be discussed, as discussed in the previous section interfaces consists of the application program. User Interface design is very important not just from user point of view but from the overall success. User Interfaces should be designed by keeping following designing issues in mind.

a)Consistency

User Interface should be consistent through out the system. The design layouts of different forms should be consistent over the whole application.

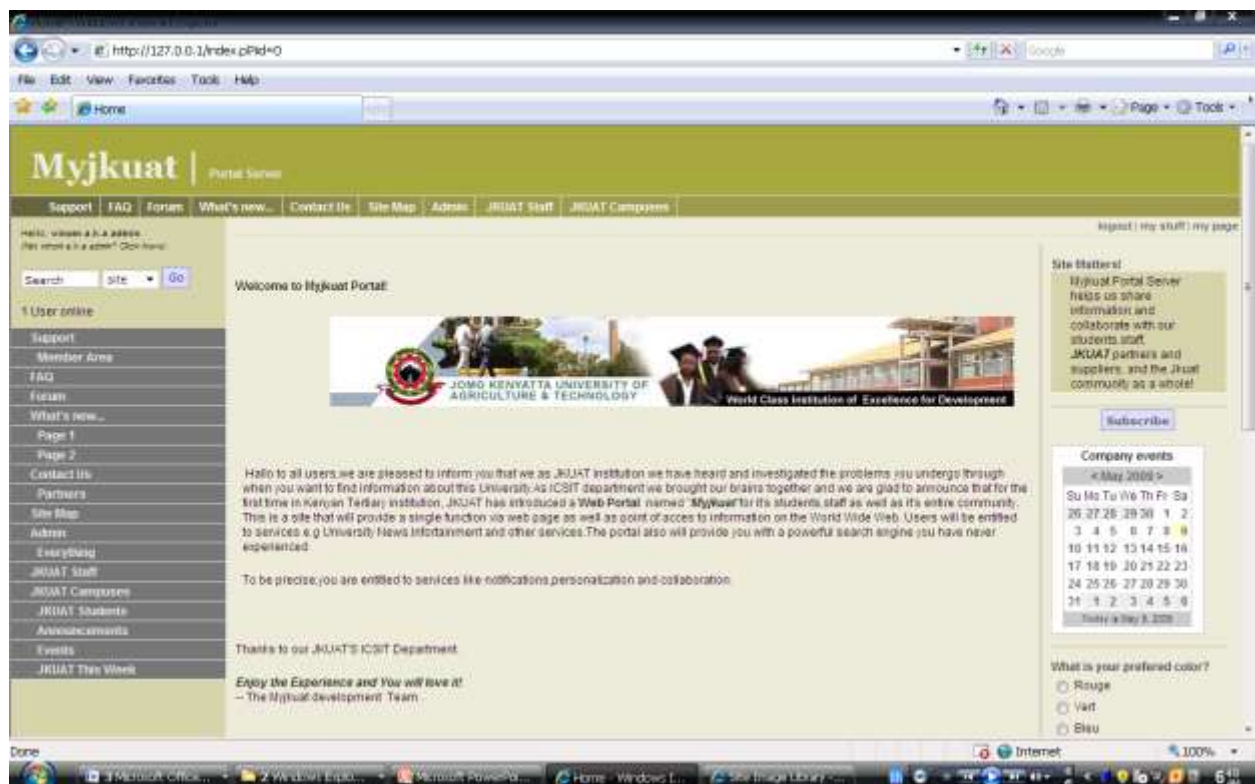
b)Prevent errors.

User interface should be designed in such way that user cannot make any serious errors. Maximum consideration should be given to place checks for invalid inputs at the user interface e.g. not allowing numeric values to entered in company name

c)Reduce short-term memory.

The human memory is another factor need to be considered while designing interfaces. Display should be kept simple, displaying of multiple pages at one instance should be consolidated.

Figure 3.1 Home page



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Figure 3.2 User Registration Form

Figure 3.3 Login Window

Figure 3.4 User Directory

4.3 Tools Used

This section describes how the system was coded. The coding was in conformance to the requirements set out in the system specification. Two development tools were used, they are; Perl framework, MySQL, HTML. Due to the capability of intergration via APIs, additional tools like Java script is also used to generate dynamic contents, message boxes, alerts and other messages.

HTML is used to to generate static pages. Apache has been used to provide server services in the web portal.

4.3.1 MySql 5.0

This tool is specifically to design and manage databases. The databases are used for storing logically related data. This tool is the core of the system for the backend purposes. It does not concern the user, however it stores all the necessary data as well as user information

4.3.2 Perl and Perl Modules

This tool is specifically for developing the application. The modules provide the built in functions that will facilitate the faster generation of special portal components.

4.3.3 Apache

This tool provides the server side services that enables clients to access applications stored in a server.

4.3.4 Internet Explorer

The tool enables the application's pages to be accessed over internet. At least I.E 5.5 and above will do. For this system am using IE 7.0.

4.4 Testing

4.4.1 Test results

The below window is an error message that is shown whenever one tries to enter wrong password or email address or both

Fig 3.5 Wrong password/Email

Fig 3.6 Registration Error Message

The figure above shows a prompt message whenever a user tries to submit an empty form

4.4.2 Evaluation of Objective Met

- i. Most of the test results met my objectives. For instance in fig 4.18, a user receives the message if he or she enters wrong email address or password. In this case our objective for authentication has been achieved.
- ii. Another incidence is on registration of users. The registration form has mandatory fields that must be filled. The user receives the above message in Fig 4.19 whenever he tries to submit uncompleted form, hence integrity of data achieved.
- iii. In this web portal, rights to access various places is based on the role of a particular user. For instance, a portal admin and a normal user have different roles, therefore you notice that the way a portal admin logs in is not the same way a normal user logs in. This will allow this different users to be directed to different pages with certain level of privileges. A portal admin will be given all the privileges. The figures bellow elaborates this.

Fig 3.7: a)Admin Login (b)Page with editing Mode

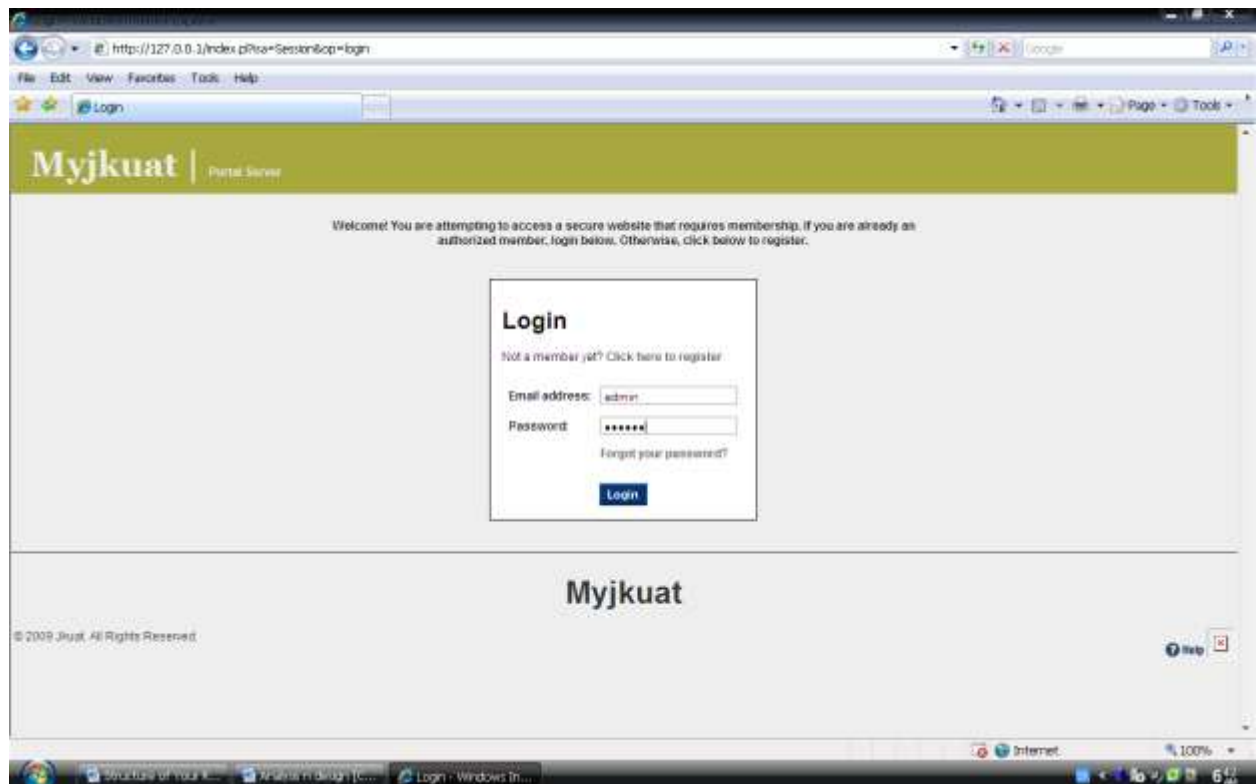


Fig 3.8 (a)Normal user login (b)Page with no privileges

4.4.3 Testing

a)Unit Testing

Unit testing focuses on the testing of the every component of the system independently. It was carried out by testing every single component of the standalone system. It was done to assure that the Database connection was working properly; SQL statements are working correctly, data reports are generated correctly.

b) Integration Testing

It focuses on the test of the system architecture and how effectively subsystem integrates with each other to perform specific function. It was done to check how well system performs with database constraints.

c)Validation Testing

Validation Testing assures that the functional requirements of the system are met, in other words validation testing helps in determining whether the proposed system has met the minimum requirements or not. The Proposed system has met all the minimum requirements that were agreed upon by the client.

d)System Testing

Finally the system testing was carried out to ensure the overall working of system was error free. System testing ensures that system works even if some component of the system comes across any problem. This can be achieved through use of good error handling mechanisms. System testing also ensures that all the mandatory fields are filled before data is stored and system shows output without any errors.

4.5 Review and Maintenance

The system will be reviewed and maintained periodically for the following reasons:

- a) To deal with unforeseen problems arising in operation.
- b) To confirm that the planned objectives are being met and to take actions if they are not.
- c) To ensure that the system is able to cope with the changing requirements of business.

Chapter 5:

5.0 Conclusion and Recommendation

The methodology employed is structured system analysis and design modeling.

The web portal has enough database for storage. I have also tried to accommodate at least all the channels required by all users. Generally the portal fulfilled all the minimum objectives that were required in order to prevent semantic attacks:

- i) Two Factor Authentication,
- ii) Role based Security.
- iii) Use of Log Events

Lastly the web portal is very powerful since I have used some of the open source features that are standardized for web portal development as well as the perl modules used.

However use of log events did not really represent what I thought I could implement. Registration blacklist required that I do research and come up with a list of email addresses that can lead to semantic attacks. I did not manage to carry out that research for such kind of addresses, furthermore such addresses are dynamic. Another weakness is that the web portal is void of photos and images which is always a mandatory to all kinds of site for usability purposes.

In order to try and minimize Semantic attacks, which is our main concern in this project, a general measure is educating portal users on odd occurrences e.g. URLs. Security is a cross-functional responsibility and groups, such as University's ICSIT, must share in the responsibility of educating students, both subordinate and non subordinate staffs and other University partners. As Internet portals increasingly become the primary gateways for conducting business, security must become a cultural principle and value for an organization to survive.

Lessons I have learned are that, developing a web portal takes a lot of time, especially when build from scratch. It is a tedious piece of work to engage into. Another issue is configuring the portal to start functioning.

I recommend that the piece of work that I have presented is not 100 percent perfect and therefore more research should be carried out to add on this. Organizations wishing to adopt the web portals, I want to encourage them to do that since web portals are powerful applications compared to websites. Incase they want to have a well customized application they should opt to develop themselves but not to start from scratch since it might take them forever to complete the application hence incurring extra costs that were not planned for.

5.1 References

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5.3 Appendices

5.3.1 Sample Codes